What the data showed

Every result and every figure, and how to read them

SRON COGNITO post-doc application

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The headline, before the detail.

A methane-trained plume detector **partially** transfers to carbon monoxide. It separates Indian steel plants from clean background at AUC 0.77; for European plants it is no better than a coin flip. Its genuine sensitivity to steel plants survives every statistical control, but is roughly a third the size of its sensitivity to how much data happens to be in the tile.

Separately, and more robustly: single overpasses mostly do not show a steel-plant plume, but stacking a site's days does — with the negative control sitting flat at zero.

THE NUMBERS

Everything on one page

The dataset

Quantity	Value
Tiles	335, from 26 sites on 18 days spread across 2019
Site-days attempted	468 — so **28% were dropped**, mostly cloud
Valid pixels per tile	81% on average
CO range	0.0222 to 0.0590 mol m^{-2} (1st to 99th percentile)
NO_2 present	333 of 335 tiles
MODIS fire present	128 of 335 tiles

The regional gradient — a sanity check that passed

Group	Median CO (mol m^{-2})	Detection rate	Stacked centre excess
Background controls	0.0290	15%	**-0.003**
European steel	0.0321	12%	+0.14
Indian cities	0.0401	27%	+0.36
Indian steel	0.0409	**52%**	**+0.59**

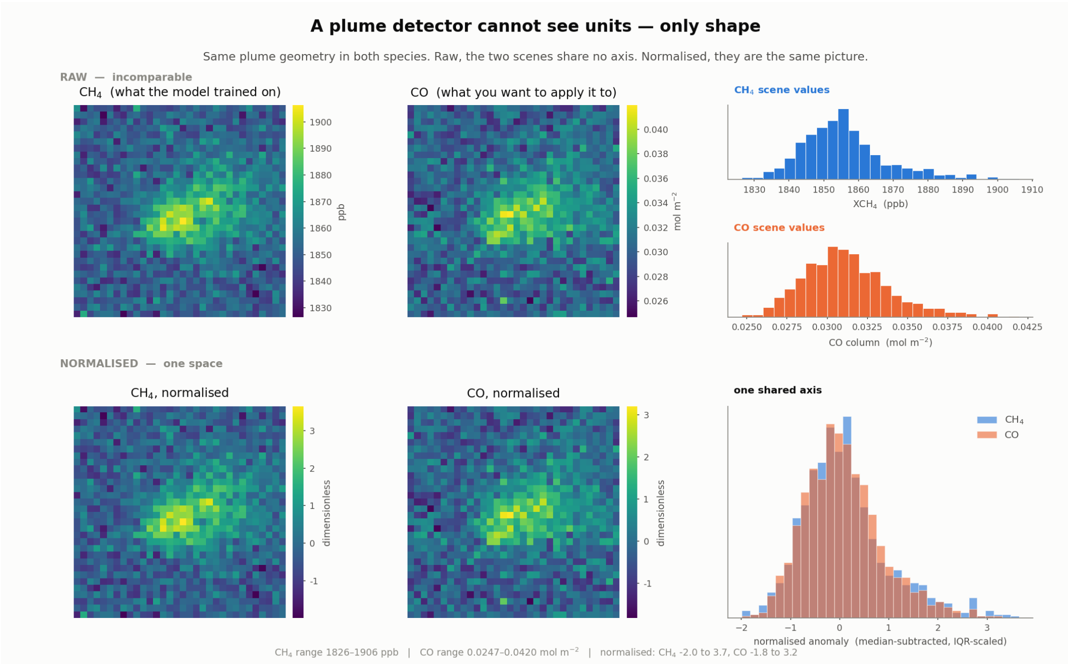
Clean marine and desert air at the bottom, the winter Indo-Gangetic Plain at the top. Nothing here is surprising, which is the point — it tells you the data is real.

The main results

Test	Result	Reading
Indian steel vs background	**AUC 0.77**	Real but modest
European steel vs background	**AUC 0.54**	Chance
Sahara control	0 of 15 detected, max 0.28	A textbook clean control
Nainital control	4 of 15 detected, max 1.00	A compromised control
Within India: steel vs non-steel	52% vs 27%, $p < 0.0001$	Not just 'India'
Dry season vs monsoon, Indian steel	59% vs 21%, $p < 0.00001$	**The strongest control**
Same test, background sites	25% vs 33%, $p = 0.73$	No swing — correct
Same test, European steel	12% vs 12%, $p = 0.46$	No swing — correct
$CO:NO_2$ fire split, India only	120 vs 177, $p = 0.003$	The fire filter works

FIGURE 1

Why a methane detector can be pointed at carbon monoxide



Top row: the same plume as methane and as carbon monoxide, raw. Bottom row: the same two after normalising.

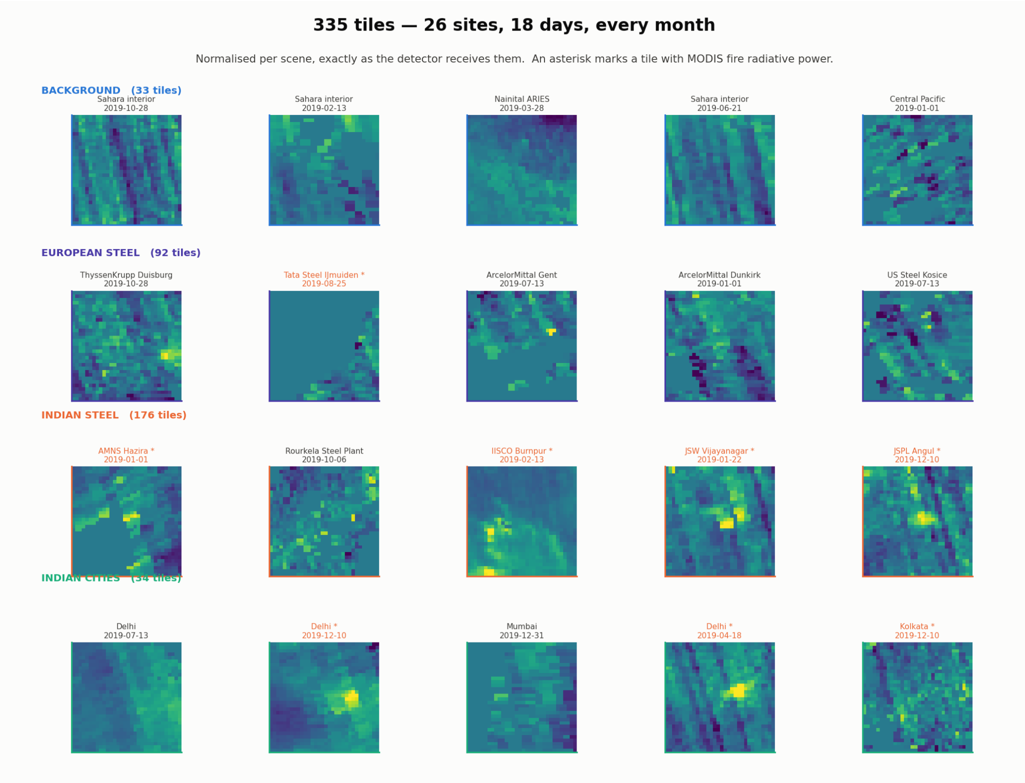
What it shows. The identical plume geometry rendered in both gases. In the top row the two scenes cannot even share an axis — methane sits near 1850 parts per billion, CO near 0.030 moles per square metre. In the bottom row, after subtracting each scene's middle value and dividing by its spread, they are statistically the same picture.

	Methane	Carbon monoxide
Plume above background	+51.6 ppb	+0.0110 mol m^{-2}
As a percentage of background	**2.8%**	**35.5%**
Peak after normalising	**3.65**	**3.20**

The asymmetry is worth raising before anyone asks. The CO plume is a far bigger fractional bump, so CO looks easier. It is not. CO's background moves around far more, and fires put plume-shaped CO almost everywhere. The difficulty moves from finding the plume to deciding what it is.

FIGURE 2

The 335 tiles, as the detector receives them



How to read it. Every tile normalised, so only shape remains. Rows are the four groups; an asterisk marks a tile where MODIS saw a fire.

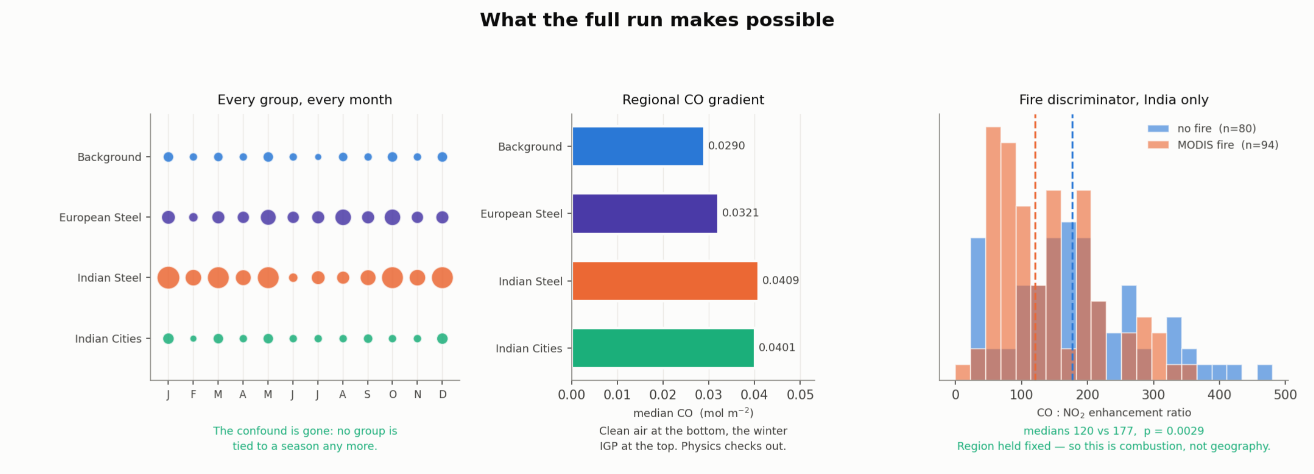
What to look for. Two textures recur and they matter:

- **Compact bright cores** sitting near the centre, sometimes with a tail. These are plumes. Bhilai and Rourkela show the clearest ones.
- **Regular diagonal banding** running corner to corner. This is the striping artefact — an instrument effect, not atmosphere. It runs along the flight track and the regridding turns it diagonal.

The second one is the important observation. A stripe is straight, elongated and coherent — the same description as a plume. This is the artefact class the published method's second stage exists to reject, and it appeared in your own data on the first day.

FIGURE 3

What the clean sampling made possible



Left — every group in every month. This is the fix for the flaw that nearly invalidated the experiment. In the previous run, European sites had been sampled only in July and August and Indian sites only in March and July, so any Europe-versus-India difference in score would have been uninterpretable. Now every group appears in all twelve months.

Middle — the regional CO gradient. Background 0.0290, Europe 0.0321, Indian cities 0.0401, Indian steel 0.0409. Physically exactly as expected, which is what tells you the pipeline is sound.

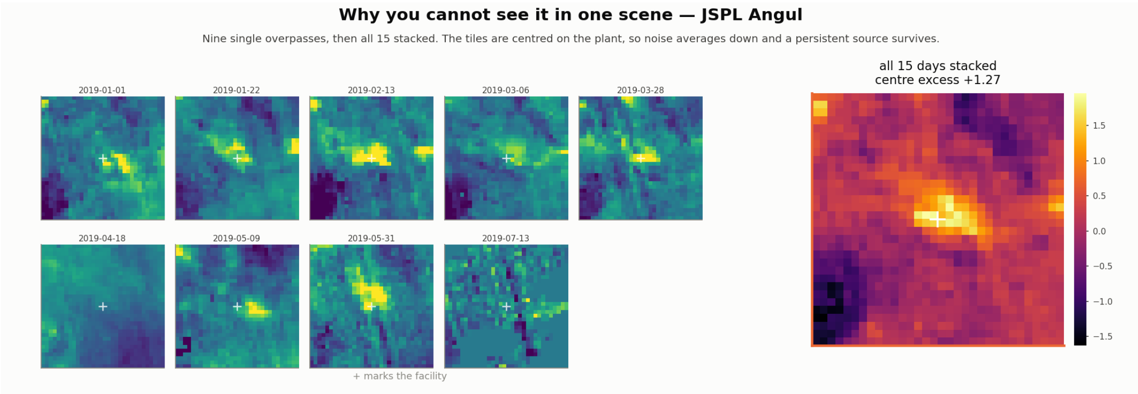
Right — the fire discriminator, India only. Scenes where MODIS saw a fire have a median $CO:NO_2$ ratio of 120; scenes without, 177. $p = 0.003$, with region held fixed so it is not geography.

> **Be precise about what the fire result means.** MODIS detects *actively flaming* fires, which burn efficiently and make relatively more NO_2 — hence the lower ratio. So this shows the ratio tracks combustion efficiency, as van der Velde's framework predicts. It does **not** by itself separate industry from fire, because furnaces also burn hot. That separation needs the ratio *plus* fire co-location.

FIGURES 4 AND 5

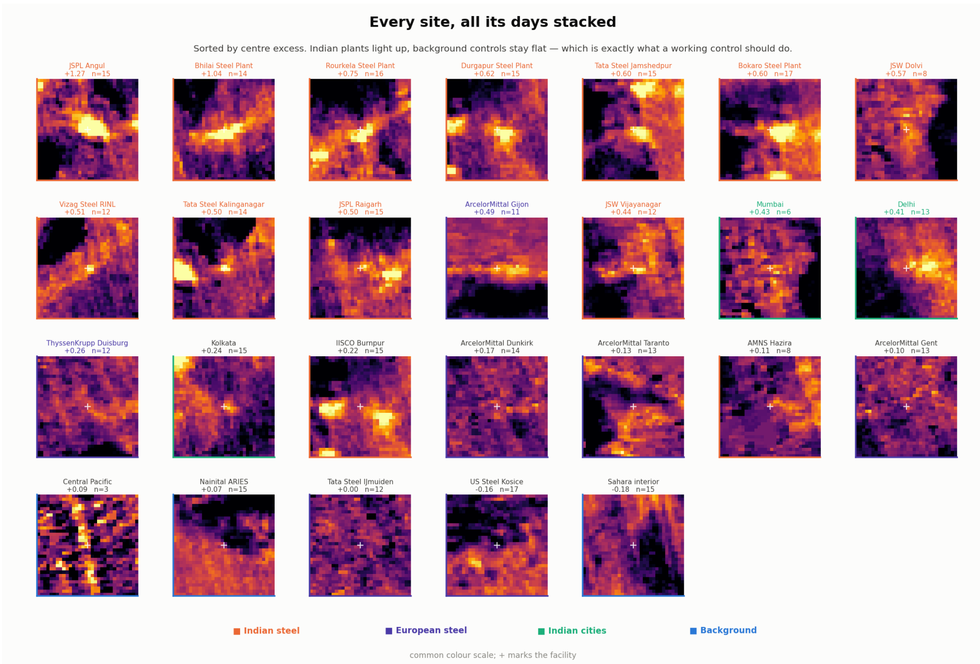
The result that does not depend on the detector at all

You said you could not see much signal at some steel sites. The number agreed: single-scene enhancement was 1.10 for background, 1.29 for Europe, 1.35 for India — heavily overlapping. **At 7 km resolution, one overpass usually does not show a steel-plant plume.**



Nine single overpasses at JSPL Angul, then all fifteen stacked.

But every tile is centred on the facility, so stacking a site's days is exactly the oversampling technique the literature uses. Noise averages down; a persistent source survives.

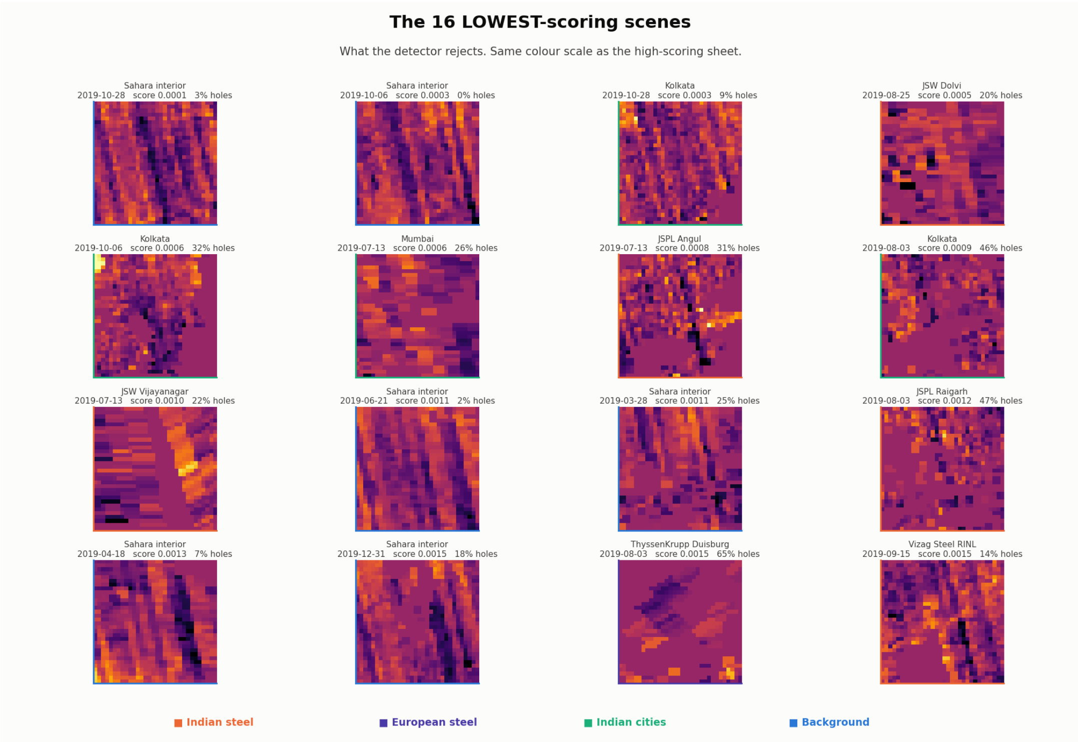


Every site, all its days stacked, sorted by centre excess. Indian plants light up; background controls stay flat.

> **Why this is arguably the best result in the project.** The background controls sit at -0.003 — flat, exactly as a working negative control must be. And the finding is substantive: steel-plant CO is not reliably a single-scene detection problem, it emerges from temporal aggregation. That is why Leguijt et al. use year-long inversions rather than per-scene detection, and it is the sharpest statement of why porting a methane super-emitter detector to CO is not straightforward.

FIGURE 6

What the detector rejects



Why this figure matters more than the high-scoring one. Anyone can show the cases where a method works. What it rejects tells you what it has actually learned.

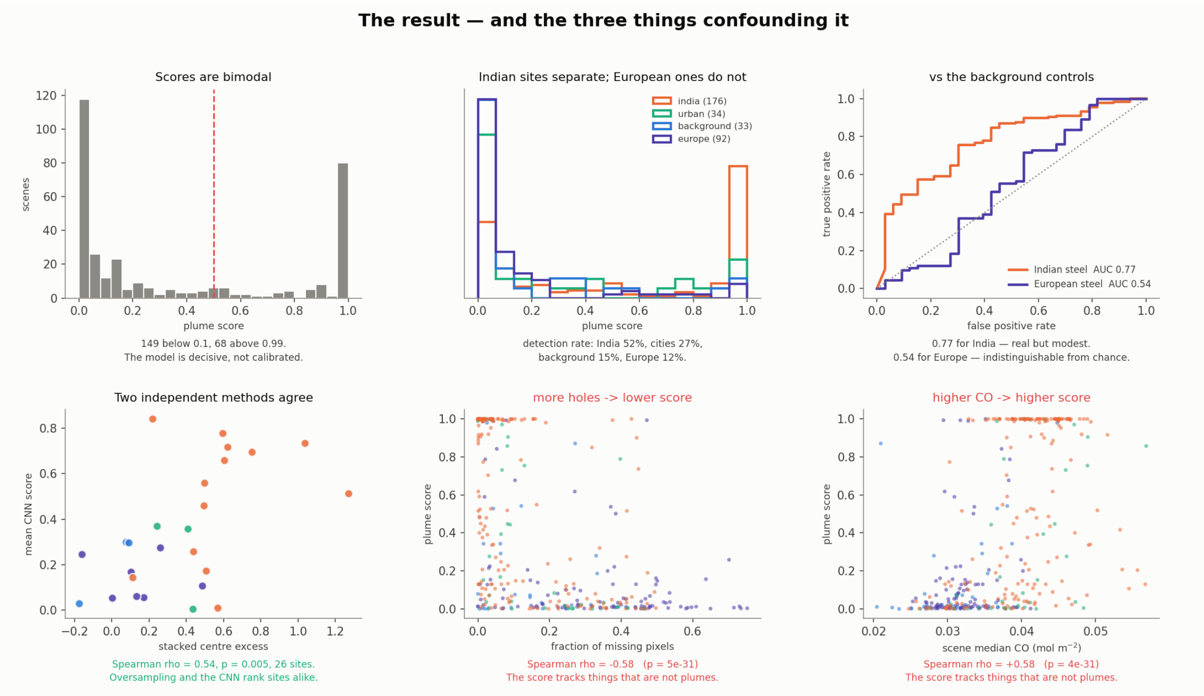
Almost every one of the sixteen lowest-scoring scenes is dominated by **diagonal banding** — Sahara at 0.0001, 0.0003, 0.0011. So the network *rejects* stripes rather than firing on them. Regular periodic banding evidently does not look like a plume to it.

This corrected a prediction I had made in the other direction, and it is worth saying in the write-up: the risk that the detector would latch onto retrieval stripes did not materialise.

The second pattern is less welcome. Several low scorers have large missing-data fractions — 65%, 47%, 46%. That is the confound quantified in the next figure.

FIGURE 7

The result, and the things confounding it



Top left — scores are bimodal. 149 tiles below 0.1, 68 above 0.99. The model is decisive, not calibrated: a score of 0.99 is not a 99% probability. Very few tiles sit near the threshold, so moving it barely changes anything.

Top middle — Indian sites separate, European ones do not. Europe's distribution is almost identical to background. In fact European steel detects at 12%, slightly *below* background's 15%.

Top right — the AUC. 0.77 for Indian steel against the background controls; 0.54 for European, which is chance.

Bottom left — a caption I had to correct. I labelled this 'two independent methods agree'. Testing it showed that holds across all 26 sites (rho 0.54, p 0.005) but **collapses within either region** — rho 0.33, p 0.27 for Indian steel alone. The agreement is the between-region contrast, not agreement on source strength. If you use this panel, the caption must change.

Bottom middle and right — the confounds. Tiles with more missing pixels score lower (rho -0.58); scenes with higher regional CO score higher (rho +0.58).

THE DECISIVE TEST

Does the signal survive the confounds?

The obvious worry is that the detector has simply learned 'India'. Two tests say otherwise.

Test one — within India

Steel plants beat non-steel Indian sites: 52% detection against 27% for cities and Nainital, median 0.53 against 0.08, $p < 0.0001$. Same region, same sampling calendar, and steel still separates.

Test two — control for everything at once

A logistic regression predicting detection, with standardised coefficients so the sizes are directly comparable:

Model	steel	median CO	missing	texture	in India
steel alone	**+0.34**				
+ regional CO	**+0.31**	+1.08			
+ missing, texture	**+0.34**	+0.72	-1.02	+0.04	
+ region	**+0.36**	+0.58	-0.99	-0.03	+0.34

Two readings, and the second is the finding:

- The steel coefficient **does not move** — +0.31 to +0.36 across every specification. It is not an artefact of the confounds.
- But it is **small**. Missing data contributes -0.99, regional CO +0.58. **The score is roughly three times more sensitive to how much data is in the tile than to whether there is a steel plant in it.**
- Texture drops to about zero once the others are included — it was never independent, just a proxy. One of the three apparent confounds dissolves under scrutiny.

Test three — the strongest control

Same coordinates, same plant, same detector. Only the weather changes.

Group	Dry season (Oct-May)	Monsoon (Jun-Sep)	p
Indian steel	**59% detected**	**21% detected**	**< 0.00001**
Indian cities	36%	0%	0.02
Nainital (no local source)	25%	33%	0.73
European steel (no monsoon)	12%	12%	0.46

Sites with a local point source in the monsoon region swing. Sites without one, or outside the monsoon region, do not. If the model were detecting 'polluted Indian atmosphere', Nainital — same region, same monsoon — would swing too.

Physically this is exactly right: monsoon convection and washout disperse a boundary-layer plume, so there is genuinely less to see.

HONESTY

What this cannot support

Every limitation below is one an expert reader would find quickly. Naming them first turns each from a weakness into evidence that you understand the method.

Limitation	Why it matters
28% of site-days were dropped	And not at random. Central Pacific lost 15 of 18 days to dark-ocean retrieval failure; June and August lost half to monsoon cloud. What survives is the clear-sky subset — inland eastern plants in the dry season, which is also where CO backgrounds are highest. The selection effect pushes the same way as the regional confound.
The seasonal test rides partly on that same effect	The monsoon sample is small (n = 29) precisely because of the cloud dropout.
The architecture is a reconstruction	Layer widths, optimiser and dropout were never published. You reproduced their approach, not their model.
The normalisation scheme is your choice, not theirs	And it is the single most consequential decision in the project.
This is gridded L2, not native L2	No averaging kernel, no retrieval diagnostics, and the quality threshold was fixed at 0.5 before you saw the data.
Detection is not quantification	Nothing here estimates an emission rate. That needs cross-sectional flux or an inversion with transport modelling — the step this project deliberately stopped short of.
Nainital is a compromised control	Himalayan terrain plus Indo-Gangetic outflow give it genuine CO structure. It detected 4 of 15. Lead with the Sahara.
Seven of twenty plant coordinates are unverified	Thirteen were checked against Global Energy Monitor; twelve were within half a pixel and JSPL Angul was 11.4 km out.

THE ANSWER

Does a methane plume detector transfer to carbon monoxide?

Neither yes nor no. A third answer, and a more useful one.

The morphology transfers. The nuisance sensitivity does not.

A methane-trained network carries over enough sense of shape to pick steel plants out of their own regional background — the effect is real, survives every control, and shows up independently in the oversampling. But its behaviour is dominated by retrieval-quality and background-level effects that were nearly absent in the clean methane scenes it learned from.

Why that is a specification, not a failure

Schuit et al.'s second stage is a classifier over 41 engineered covariates — quality value, cloud fraction, albedo, surface pressure. Those are precisely the quantities that would absorb the two dominant terms measured here and leave the plume term standing.

You independently derived the requirement for that stage, with coefficients attached. That is a more sophisticated outcome than a good AUC would have been.

The European null is a result in its own right

European plants score 12%, below background's 15%, with a stacked excess of +0.14 against the control's -0.003. Yet Leguijt et al. successfully quantified those same plants using year-long analytical inversions on the same instrument in the same year.

So: **per-scene detection fails where annual inversion succeeds.** That is a clean statement about the limits of a detection-first approach for CO, and it is directly relevant to how COGNITO should be built.

One sentence for the motivation letter

To test whether the detection approach transfers across species, I reproduced the neural-network stage of Schuit et al. (2023) on your published training scenes and applied it to TROPOMI CO columns over European and Indian steel plants. The transfer is partial — AUC 0.77 against clean background for Indian plants, chance for European ones — and the residual sensitivity is dominated by retrieval-quality and background-level covariates, which is a quantitative argument for the artefact classifier in your stage two.